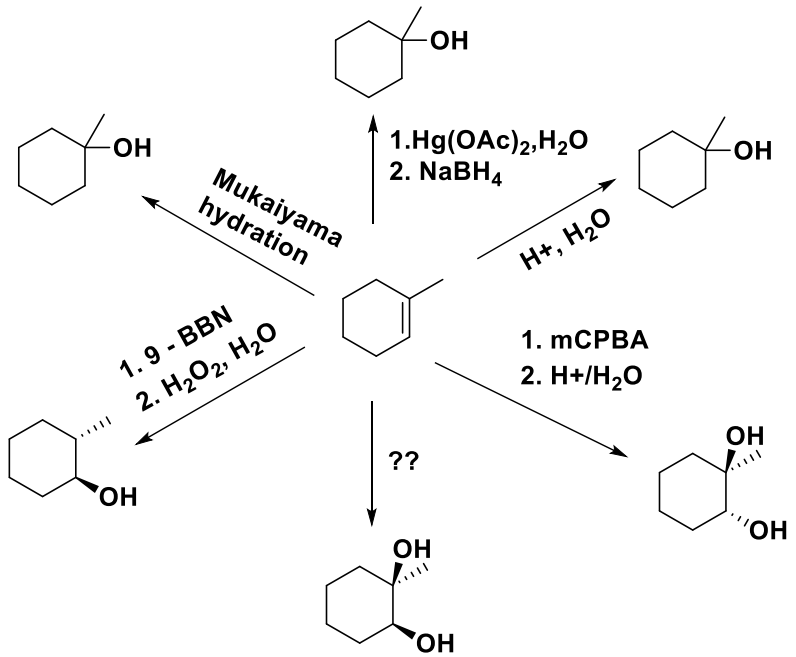
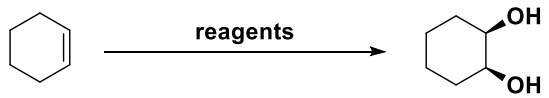
Summery of oxidation of alkenes



Dihydroxylation Of Alkene :



Reagents:

A. $KMnO_4$, ^-OH, H_2O, low temp.

or,

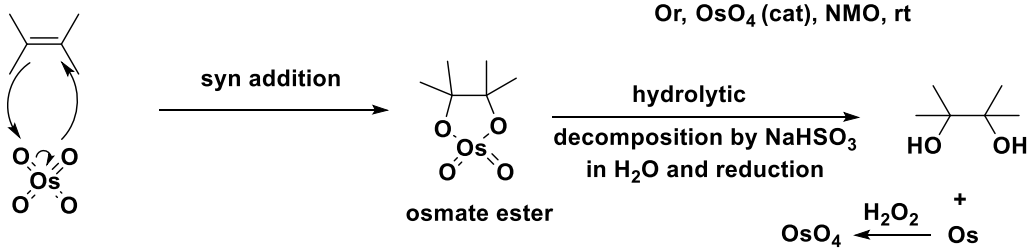
B. i. OsO_4, pyridine

ii. $NaHSO_3$ / H_2O

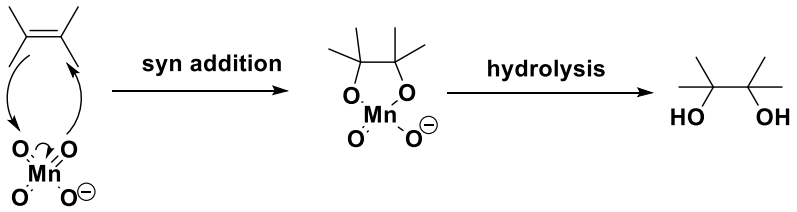
or, Na_2SO_3, H_2O

Or, OsO_4 (cat), NMO, rt

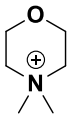
Mechanism :



same mechanism for MnO_4^-

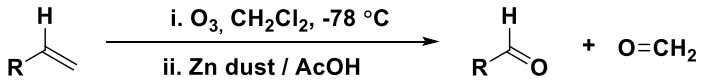


OsO4 is the most widely used.
- highly toxic, expensive, volatile
- catalytic method.

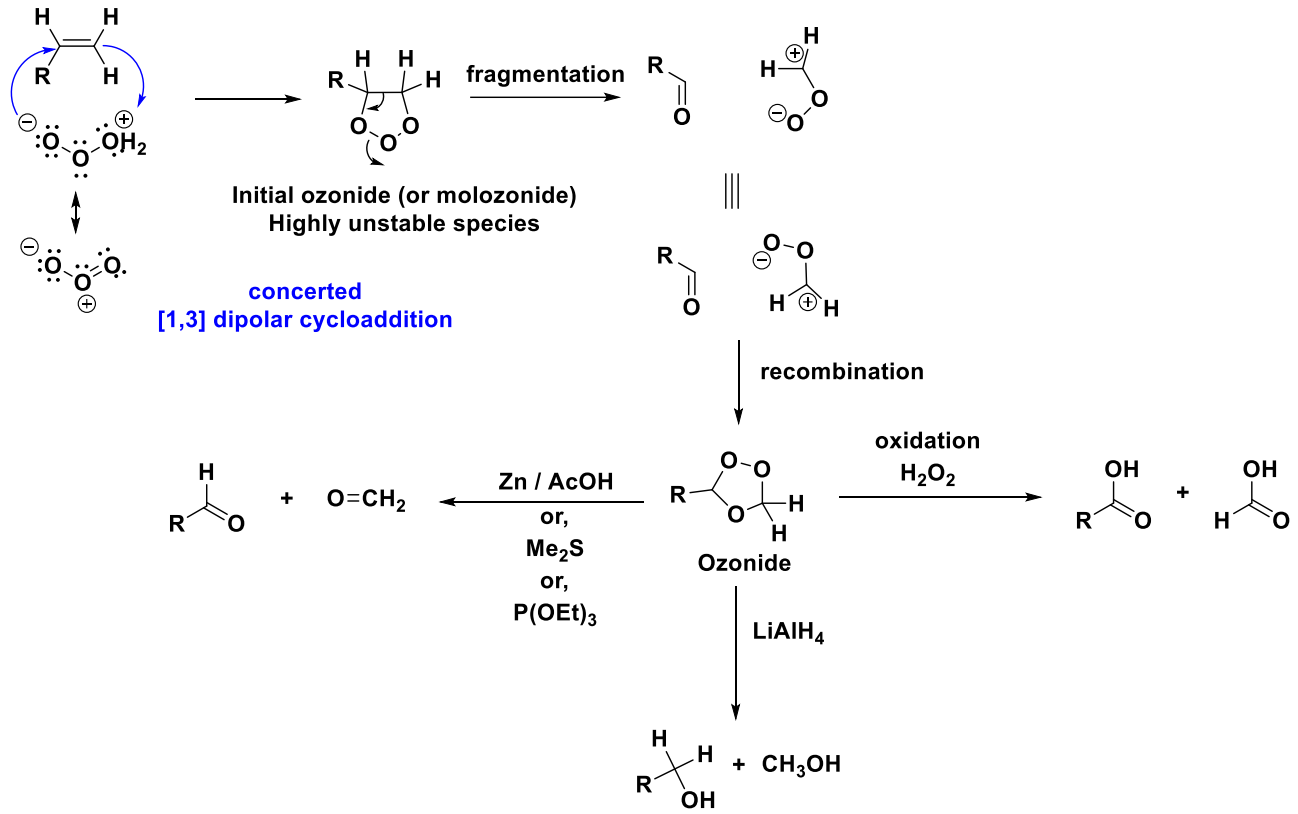


N - methylmorpholine-N-oxide is used as co-oxidant

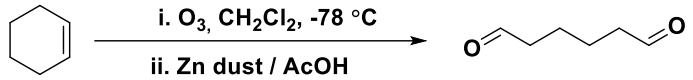
Ozonolysis of alkene :



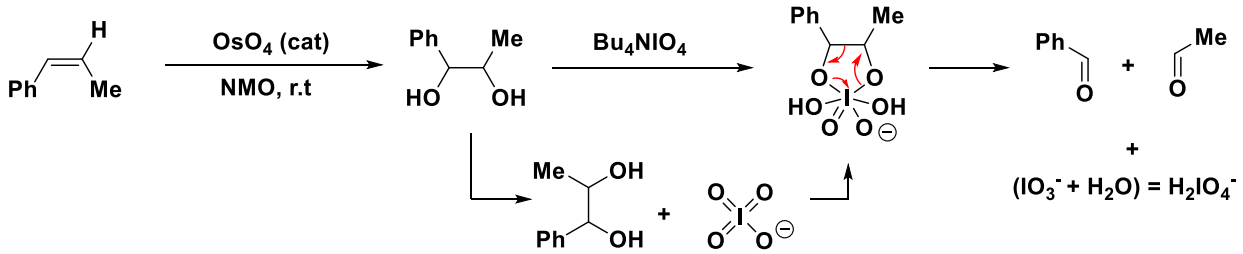
Mechanism :



Example:



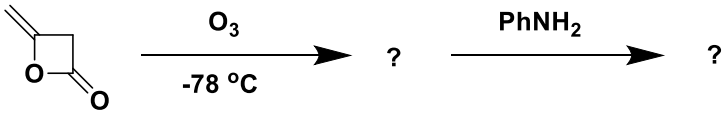
Other method for oxidative cleavage



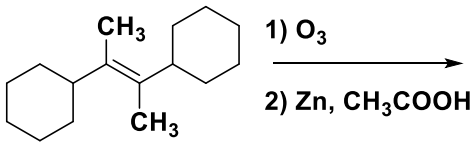
Questions:

Write major products with proper mechanism.

A.



B.



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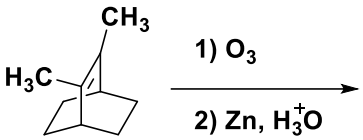

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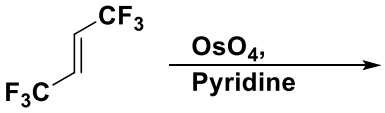
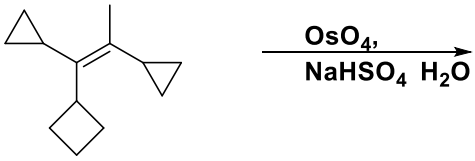
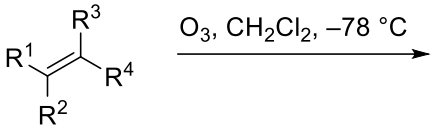


E.



2. **Write the structure of possible ozonides for the following reaction**